SEC P vs NP log 2026-05-07

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 14:30 UTC

SEC P vs NP — daily log 2026-05-07

30 cycles recorded

Discrete Morse Critical-Cell Count Lower Bounds Frege Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete Morse Theory (Forman combinatorial topology) × Tree-like Frege / Resolution proof DAG depth (gateway to EF lower bounds)
- **Recorded**: 2026-05-07 18:59 UTC
- **Entry ID**: `cac9d682890f`

Statement

Let F be an unsatisfiable 3-CNF on n variables, and let π be the resolution refutation produced by a fixed DPLL strategy (JW-OS branching, no learning); π lifts to a Frege proof of ¬F whose depth d(π) is the longest antecedent chain. View the proof DAG G_π=(V,E) as a 1-dimensional simplicial complex and apply the canonical greedy Forman matching: process vertices in reverse topological order, pairing each unmatched vertex with its lowest-index unmatched incoming edge; let M(F) be the resulting count of critical 1-cells. Conjecture (C): for every unsatisfiable 3-CNF F with $n \le 40$, $M(F) \ge d(\pi) - \lceil \log_2 |V| \rceil$, and the asymptotic mean over random 3-CNF at ratio 4.5 satisfies $E[M(F)] = \Theta(n / \log n)$; a single instance violating the per-instance inequality refutes C.

Rationale

Forman's discrete Morse theory compresses simplicial complexes while preserving homotopy type, and critical 1-cells encode homologically essential branching that cannot be matched away. Proof DAGs have a natural 1-complex structure, but discrete Morse theory has essentially never been wired to propositional proof complexity (<5 papers, none on EF/Frege depth). If C holds, every short Frege proof must have a greedy Morse matching that nearly perfectly pairs lines with edges, giving a new combinatorial obstruction to EF compactness for hard tautologies.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [2504.15729v2] Strong discrete Morse theory - [0805.2131v2] Functoriality in Morse theory on closed manifolds - [0406486v2] The Morse Complex for a Morse Function on a Manifold with Corners - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [2111.07483v2] Tradeoffs for small-depth Frege proofs

Empirical Test

- exit code: 1, elapsed: 0.62s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92db060e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92db060e.py", line 105, in run_trial
    if not dpll(cnf, assignment):
           ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92db060e.py", line 37, in dpll
    if all(lit not in assignment or assignment[lit] != val for lit, val in clause):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92db060e.py", line 37, in <genexpr>
    if all(lit not in assignment or assignment[lit] != val for lit, val in clause):
                                                               ^^^^^^^^
TypeError: cannot unpack non-iterable int object
```

Judge reasoning

The test crashed with a TypeError during DPLL execution (clause unpacking failure), so no data was produced and the pre-registered support condition could not be evaluated. | next: Fix the clause representation in the DPLL routine (ensure clauses are iterables of (lit,val) tuples, not bare ints) and rerun the full n∈{10,...,40} sweep.

Additive Energy Threshold and Deterministic Communication Complexity

- **Verdict**: INCONCLUSIVE
- **Bridge**: Additive combinatorics × Deterministic communication complexity
- **Recorded**: 2026-05-07 19:01 UTC
- **Entry ID**: ec6a507b1136

Statement

For any Boolean function f on n variables, if the additive energy of its truth table $\Lambda(f) \geq n^3$, then the deterministic communication complexity of f is $\Omega(n)$.

Rationale

High additive energy implies a rich additive structure, which may necessitate high communication complexity due to the need for resolving conflicting additive relations between inputs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.41s

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of support/falsification criteria. | next: Run test with increased time limit and targeted parameter adjustments for critical cases

---

Cheeger Constant Lower Bound on Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Proof Length
- **Recorded**: 2026-05-07 19:11 UTC
- **Entry ID**: `9dd1a19fd8df`

Statement

For any connected graph G with n vertices, the resolution proof length of the Tseitin formula derived from G is at least 2^Ω(h(G)), where h(G) is the Cheeger constant of G.

Rationale

The Cheeger constant captures the expansion properties of G, which are known to influence the complexity of Tseitin formulas. A higher Cheeger constant (better expansion) implies longer resolution proofs, as expanders require exponential proof lengths.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c326aad.py", line 93, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c326aad.py", line 77, in run_trial
    proof_length = len(tseitin_clauses) if dpll_solve(tseitin_clauses) else float('inf')
                                           ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c326aad.py", line 71, in dpll_solve
    return dpll(model, clauses)
           ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c326aad.py", line 62, in dpll
    new_clause = [l for l, v in zip(clause, [True, False]) if new_model[l] != v]
                                                              ~~~~~~~~~^^^
KeyError: '-p24'
```

Judge reasoning

Test crashed before producing data, preventing evaluation of conjecture support or falsification. | next: Debug and rerun the test with error handling for clause variables

---

Sandpile 2-Rank Lower-Bounds Tree-Resolution Size of Tseitin

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Chip-firing / Sandpile (critical) groups of graphs × Tree-resolution refutation size (sub-Frege; gateway to Frege depth lower bounds)
- **Recorded**: 2026-05-07 19:28 UTC
- **Entry ID**: `82f0b7df0af9`

Statement

Let G=(V,E) be a connected graph with odd-parity charge c:V→F_2, and let T(G,c) be the (unsatisfiable) Tseitin CNF. Let $K(G)=Z^{\{|V|-1\}/\tilde{L}(G)Z}$ {|V|-1} denote the sandpile group of G, computed from the reduced Laplacian $\tilde{L}(G)$ (delete one row/column of the combinatorial Laplacian) via Smith Normal Form yielding invariant factors s_1|s_2|…|s_{|V|-1}; define the dyadic rank r_2(G)=#{i: s_i is even}. Then the minimum tree-resolution refutation size satisfies size_TR(T(G,c)) ≥ |V|·2^{r_2(G)}, and a single graph G with size_TR(T(G,c)) < |V|·2^{r_2(G)} refutes the conjecture.

Rationale

The sandpile group encodes global cycle/flow structure of G as integer torsion, and its 2-Sylow part precisely measures independent F_2-cycle classes — the same algebraic obstruction that makes Tseitin unsatisfiable. Heuristically each dyadic invariant factor furnishes an independent parity certificate that a tree refutation must dispatch via an additional case-split, doubling the tree size. Chip-firing has been linked to spanning-tree counts and graph Riemann-Roch but, to our knowledge, never to Tseitin proof size, providing a virgin algebraic-combinatorial bridge to Frege-style lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `2.20s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ee706686.py", line 187, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ee706686.py", line 155, in run_trial
    cnf = tseitin_cnf(graph, n, charge)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ee706686.py", line 54, in tseitin_cnf
    clause = [literals[i][charge[i]]]
              ~~~~~~~~~~~^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in tseitin_cnf before producing any data, so neither the support nor falsification condition can be evaluated. | next: Fix the indexing bug in tseitin_cnf (likely a mismatch between vertex count and the literals/charge arrays) and rerun the 720-instance pre-registered protocol.

Real Stable Polynomial Coefficient Sum and AC^0 PARITY Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuits Computing PARITY
- **Recorded**: 2026-05-07 19:29 UTC
- **Entry ID**: `3dd7d5ee43a7`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the sum of absolute values of coefficients of the real stable polynomial representing C is ≥ $2^{\{\Omega(n}$\{1/d\})\}, where d is the depth of C.

Rationale

Real stable polynomials encode positivity constraints that mirror AC^0 circuit structure. Their coefficient sums reflect combinatorial complexity, with exponential growth implying depth-limited size lower bounds for PARITY.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.67s`

```
: False, 'counterexample': 'coeff_sum=0, threshold=1.1904332152525279'}
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 0, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 0, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 2, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 1, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'coefficient_sum', 'metric_value': 2, 'instances_tested': 1, 'conjecture_hold
RESULT: FALSIFIED counterexample="coeff_sum=1, threshold=1.186425250131506" first_failing_seed=23
```

Judge reasoning

The coefficient_sum metric is saturated at ≤2 across all trials, failing to demonstrate exponential growth required by the conjecture. Test instances likely use n=1 or shallow depths, preventing observation of scaling behavior. | next: Test with larger n and deeper AC^0 circuits to observe scaling of coefficient_sum

---

Additive Energy Gap Between P and AC^0 Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × Boolean Circuit Complexity
- **Recorded**: 2026-05-07 19:41 UTC
- **Entry ID**: `735ee8b65843`

Statement

For any explicit Boolean function $f \in P$ with n variables, its additive energy $E(f) = |\{(x,y,z,w) \in \{0,1\}^{n^4} : f(x) + f(y) = f(z) + f(w)\}|$ satisfies $E(f) \leq O(n^2)$. Conversely, for any AC^0 circuit computing f, $E(f) \geq \Omega(2^{2n}/n^2)$.

Rationale

Additive energy captures the structured regularity of a function's truth table. AC^0 circuits, with their bounded-depth structure, inherently exhibit higher additive energy due to correlation with low-degree polynomials, while explicit P-functions (e.g., parity, modular counting) have pseudorandom truth tables with sparse additive structure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase time limit and re-run test with stricter resource constraints

---

Spectral Mahler Measure of Tseitin Moment Matrix Bounds SOS Refutation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Mahler measure / logarithmic heights of integer-spectrum matrices (analytic-Diophantine number theory) × Degree-2 SOS moment matrix for Tseitin Max-CUT on regular expanders
- **Recorded**: 2026-05-07 20:05 UTC
- **Entry ID**: `146b8cf9b370`

Statement

Let $T(G,b)$ be an unsatisfiable Tseitin formula on a connected d-regular graph G ($d \geq 3$) with parity charges $b \in \{0,1\}^V$, and let $A(G,b)$ be the signed adjacency matrix with $A_{ij}=(-1)^{b_i+b_j}$ if $\{i,j\} \in E$ and 0 otherwise. Form the $(n+1)\times(n+1)$ canonical degree-2 SOS moment matrix $M(G,b)$ by setting $M_{0,0}=1$, $M_{0,i}=0$, $M_{i,i}=1$, and $M_{i,j}=A_{ij}/(d-1)$ for $i,j \geq 1$; define the spectral Mahler measure $m(M)=\Sigma_{i:|\lambda_i|>1} \log|\lambda_i|$ over eigenvalues of M. Conjecture: for every $(d,2\sqrt{d-1})$-spectral expander G and every parity-violating b ($\Sigma b_v \equiv 1 \bmod 2$), $m(M(G,b)) \geq \alpha_d \cdot n - O(\log n)$ with $\alpha_d = (1/2)\cdot\log((d-1)/(d-2)) > 0$, and any degree-2 pseudoexpectation certifying a $(0.878+\varepsilon)$-approximation to Max-CUT on G must have $m(M)$ strictly below $\alpha_d \cdot n/2$, so its existence is forbidden whenever the conjecture holds.

Rationale

Tseitin lower bounds against algebraic relaxations are typically proved via spectral expansion of an associated matrix, but spectral norm alone discards the multiplicative information encoded in the full eigenvalue sequence; the Mahler measure $m(M)=\log\prod\max(1,|\lambda_i|)$ is precisely the multiplicative invariant from Diophantine geometry (Lehmer's problem, height theory) that aggregates this information. Linking this height-theoretic invariant to SOS feasibility on Tseitin Max-CUT bridges a number-theoretic field essentially absent from proof complexity (▯5 prior papers) with the canonical SOS-vs-Tseitin separation question. Because m(M) is a local function of the spectrum it neither algebrizes (it is not preserved under oracle relativization of M) nor falls to natural proofs (it is a property of the SOS object, not of a Boolean function family).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.12s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing any data, so none of the pre-registered support conditions could be evaluated. The critic also challenges, and no metric is available to confirm or refute the bound. | next: Optimize the eigenvalue computation (e.g., use sparse SciPy eigensolvers and reduce seed count or n-range) and rerun with a higher timeout to obtain the m(M)/n statistics.

Polymatroid Rank Gap in Monotone DNF Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone Circuit Depth for k-CLIQUE
- **Recorded**: 2026-05-07 20:16 UTC
- **Entry ID**: `e3302ada2f9d`

Statement

For any k-CLIQUE indicator function f on n vertices, the polymatroid rank function ρ_f of its hypergraph representation satisfies $\rho_f \geq \Omega(n/k)$ while remaining $\leq O(\log n)$ for all DNFs with size poly(n).

Rationale

Polymatroid rank captures submodular structure of hypergraphs; its gap between k-CLIQUE and general DNF could expose inherent depth complexity. The conjecture links combinatorial optimization (polymatroids) to monotone circuit depth, bypassing traditional algebraic barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dbf79bba.py", line 89, in <module>
    result = run_trial(seed)
```

```
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dbf79bba.py", line 65, in run_trial
    rank_f = hypergraph_rank([(abs(lit), lit // abs(lit)) for clause in formula for lit in clause], n)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dbf79bba.py", line 37, in hypergraph_ra
    if rank[u] == 0 or rank[v] == 0:
       ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before collecting data; no results available to assess conjecture | next: Fix hypergraph_rank implementation to handle empty lists/indices properly

---

Hashimoto Non-Backtracking Pole Gap Lower-Bounds Tseitin Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ihara zeta functions of finite graphs (number-theoretic non-backtracking spectra, Bass–Hashimoto theory) × Resolution refutation length of Tseitin formulas T(G,σ)
- **Recorded**: 2026-05-07 20:33 UTC
- **Entry ID**: `741fc0a4ec69`

Statement

Let G=(V,E) be any connected graph (possibly irregular) and B_G be its 2|E|×2|E| Hashimoto non-backtracking operator (B_{(a,b),(c,d)}=1 iff b=c and a≠d). Define the Ihara pole gap ν(G):= log|λ_1(B_G)| − log max{|λ|: λ∈Spec(B_G), λ∉{±λ_1,±1}}, set to 0 if no such second eigenvalue exists. Then for every odd-charge labeling σ:V→{0,1} with Σσ(v)≡1 (mod 2), the minimum Resolution refutation length L(T(G,σ)) satisfies log_2 L(T(G,σ)) ≥ c·|V|·ν(G) for an absolute constant c>0, and ν(G)=O(1/|V|) on every graph admitting an O(1)-vertex separator.

Rationale

Standard expansion arguments use λ_2 of the adjacency or Laplacian matrix, but the Bass–Hashimoto operator captures non-backtracking closed-walk multiplicities that better reflect how Resolution must trace clause-derivation paths without revisiting variables. Because B_G is the natural transfer operator of the Ihara zeta function, its second-pole gap is the right quantitative analogue of a 'prime number theorem' for graphs — and bottleneck graphs (long paths, dumbbells, weak cuts) automatically have collapsed gap. Importing zeta-function multiplicities into proof complexity is essentially uncharted (<5 papers) and avoids the algebrization barrier because B_G is a combinatorial integer matrix derived purely from the formula's underlying graph.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0311029v1] Zeta functions over zeros of general zeta and L-functions - [1306.5454v1] The Ihara zeta function of the infinite grid - [1712.05169v12] Zeros and the functional equation of the quadrilateral zeta function - [2112.11882v2] Explicit values for Ramanujan's theta function $\varphi(q)$ - [0904.1226v1] On an Asymptotic Series of Ramanujan

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_baef02fb.py", line 120, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_baef02fb.py", line 69, in run_trial
    B_G = hashimoto_operator(G)
          ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_baef02fb.py", line 24, in hashimoto_ope
    for (a, b), (c, d) in E:
        ^^^^^^
TypeError: cannot unpack non-iterable int object
```

Judge reasoning

The test crashed with a TypeError in hashimoto_operator before any data was produced, so the pre-registered regression and per-instance criteria cannot be evaluated. No RESULT line was emitted and the critic challenged on the same grounds. | next: Fix the Hashimoto operator construction to iterate over directed edge pairs as tuples (e.g., build E as a list of (u,v) tuples and index with enumerate) and rerun the full 1080-instance protocol.

Tropical Convex Hull Dimension and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Convex Geometry × ACC^0 Circuit Size
- **Recorded**: 2026-05-07 20:41 UTC
- **Entry ID**: `9e0cac4cfcf9`

Statement

For any CNF formula Φ with n variables, the dimension of the tropical convex hull of its clause vectors is Θ(log n) if and only if Φ has an ACC^0 circuit of size Θ(n).

Rationale

Tropical convex hulls capture the extremal structure of clause interactions, which may constrain the algebraic complexity of ACC^0 circuits. The logarithmic scaling suggests a trade-off between geometric sparsity and computational efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.12s`


```
'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 14, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 50, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 46, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 18, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 76, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 32, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 72, 'instances_tested': 1, 'conjecture_hold
```

```
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 38, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 34, 'instances_tested': 1, 'conjecture_hold

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f6948cba.py", line 74, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f6948cba.py", line 74, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed with KeyError 'seed' before producing valid data, preventing reliable assessment of conjecture status. | next: Fix test to handle 'seed' parameter properly and re-run with validated input configurations

---

Fourier Coefficient Sum Lower-Bounds Resolution Length for 3-CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis of Boolean Functions × Resolution Proof Length
- **Recorded**: 2026-05-07 20:48 UTC
- **Entry ID**: `e2aedb4aa03f`

Statement

For every 3-CNF formula Φ with n variables, the resolution proof length is at least Σ_{S⊆[n]} |f̂(S)|, where f̂(S) is the Fourier coefficient of the characteristic function of Φ. Equality holds when Φ is a single clause.

Rationale

Fourier coefficients capture the sensitivity of boolean functions to variable flips, which directly relates to the number of resolution steps needed to eliminate clauses. The sum of absolute coefficients quantifies the overall 'spread' of the function's structure, which must be addressed in any refutation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.41s`

```
e': 'Fourier Coefficient Sum', 'metric_value': 17, 'instances_tested': 1024, 'conjecture_holds': True
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 35, 'instances_tested': 1024, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 23, 'instances_tested': 1024, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 16, 'instances_tested': 1024, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 18, 'instances_tested': 1024, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 26, 'instances_tested': 1024, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 15, 'instances_tested': 1024, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 23, 'instances_tested': 1024, 'conj
```

```
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 26, 'instances_tested': 1024, 'conj

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d64c59.py", line 85, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d64c59.py", line 85, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                           ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed due to KeyError 'seed' before completing required data collection | next: Fix test to handle seed tracking and re-run with sufficient sample size

---

Non-Abelian Fourier Coefficient Boundedness in Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-Abelian Harmonic Analysis × Resolution Proof Length
- **Recorded**: 2026-05-07 20:58 UTC
- **Entry ID**: `226cd77ad310`

Statement

For every Tseitin formula T(G, σ) on an expander graph G with n vertices, the resolution proof length is bounded by O(μ(G)), where μ(G) is the maximum absolute value of the non-abelian Fourier coefficients of G's adjacency matrix over its automorphism group. This bound is tight up to polylog(n) factors.

Rationale

Expander graphs' spectral properties, captured via non-abelian harmonic analysis, directly influence the combinatorial complexity of refuting Tseitin formulas. The Fourier coefficients quantify the non-uniformity of the graph's expansion, which correlates with the required proof size in resolution.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.08s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run with stricter resource limits to capture full execution data

---

Free Cumulant κ_4 of Disjointness Spectrum Lower-Bounds R(DISJ)

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free probability (Voiculescu free cumulants of empirical singular-value distributions) × SoS degree-2 moment matrix view of randomized communication complexity of DISJOINTNESS
- **Recorded**: 2026-05-07 21:02 UTC
- **Entry ID**: `c30c16c159e9`

Statement

For k ≥ 2 let Π_k ∈ {−1,+1}$^{\{2}$k×2^k} be the sign-disjointness matrix Π_k[x,y] = (−1)^{[x∧y ≠ 0]}, and let μ_k be the empirical distribution of the squared singular values of Π_k/√(N·‖Π_k‖_F²/N) (so m_1(μ_k)=1, N=2^k). Define the fourth free cumulant $\kappa_4(\mu) := m_4(\mu) - 2 \cdot m_2(\mu)^2$. We conjecture (i) κ_4(μ_k) ≥ c·k for an absolute constant c > 0, while (ii) for an N×N uniformly random ±1 sign matrix R, E[κ_4(μ_R)] = O(1/N); equivalently, the κ_4 of the SoS-degree-2 moment matrix of any sign matrix M furnishes a lower bound R^pub(M) ≥ Ω(κ_4(μ_M)) on randomized public-coin communication complexity, and DISJ saturates this bound to Ω(k).

Rationale

Random sign matrices have semicircular limiting spectra whose free cumulants vanish above order 2, so κ_4 detects deviation from freeness — exactly the structured 'tensor-power' rigidity that makes DISJ hard. Because Π_k = M^{⊗k} for the 2×2 sign block M, its squared spectrum is the k-fold classical convolution of a non-semicircular law, forcing κ_4 to grow linearly in k under the multiplicativity of moments. Free cumulants are read directly off the SoS degree-2 pseudoexpectation moment matrix, giving a clean SoS-hierarchy invariant that is unrelativizing (depends on actual matrix entries) and avoids algebrization (no oracle algebraic extension can fake non-semicircular spectra of explicit tensor-power matrices).

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2207.14810v3] Simplifying a classical-quantum algorithm interpolation with quantum singular value transformations - [2303.00713v4] Full Eigenstate Thermalization via Free Cumulants in Quantum Lattice Systems - [2603.19490v1] Communication Complexity of Disjointness under Product Distributions - [1608.06580v2] Communication complexity of approximate Nash equilibria - [1210.1461v1] A Scalable CUR Matrix Decomposition Algorithm: Lower Time Complexity and Tighter Bound

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_76f8276b.py", line 95, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_76f8276b.py", line 69, in run_trial
    sv = singular_values(Pi_k)
         ^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_76f8276b.py", line 56, in singular_valu
    U, _, Vt = gaussian_elimination(matrix_multiply(A, A))
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_76f8276b.py", line 27, in gaussian_elim
    factor = A[j][i] / A[i][i]
```

```
             ~~~~~~~~^~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError in a hand-rolled gaussian_elimination routine before any κ_4 values were computed, so neither the support nor falsification conditions can be evaluated. | next: Replace the custom SVD with numpy.linalg.svd (or eigvalsh on AA^T) to robustly compute singular values of Π_k and random sign matrices, then re-run the pre-registered criterion across k∈{2..6}.

Algebraic Matroid Rank and ACC^0 Circuit Size Trade-off

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Matroid Theory × ACC^0 Circuit Size
- **Recorded**: 2026-05-07 21:03 UTC
- **Entry ID**: `ffcf1be9a920`

Statement

For any CNF formula Φ with n variables, the rank of its incidence matrix's algebraic matroid over GF(2) satisfies rank(Φ) ≥ log n if and only if Φ requires ACC^0 circuits of size Ω(n^c) for some constant c ≥ 2.

Rationale

Algebraic matroid rank captures linear dependencies in clause-variable incidence, which may constrain the parallelism of ACC^0 circuits. High rank implies non-trivial algebraic structure, potentially forcing larger circuit sizes via combinatorial expansion.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4c227f8a.py", line 106, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4c227f8a.py", line 57, in run_trial
    rank = gaussian_elimination(incidence)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4c227f8a.py", line 27, in gaussian_elim
    factor = Fraction(matrix[j][i], matrix[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
   File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing reliable evaluation of the conjecture's validity. | next: Debug Gaussian elimination implementation to handle singular matrices and re-run tests

---

Maximum Fourier Coefficient Lower Bound via Sensitivity Scaling

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis of Boolean Functions × Decision Tree Size
- **Recorded**: 2026-05-07 21:18 UTC
- **Entry ID**: `eff6dec1e004`

Statement

For all boolean functions f: {0,1}^n → {0,1}, max_{S} |f̂(S)| ≥ sensitivity(f) / √n. Equality holds for parity functions.

Rationale

Fourier coefficients encode variable influence, while sensitivity measures local changes. This conjecture bridges their relationship, suggesting that high sensitivity implies significant Fourier mass, which could expose structural constraints on decision tree size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.30s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing verification of support fraction or counterexamples. | next: Run test with extended timeout and debug crash logs

---

Roth Density of ACC^0 Truth Tables Lower-Bounds Sipser Hardness

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive combinatorics: Roth-type 3-AP density in Z_{2^n} under binary integer encoding × $ACC^0[6]$ circuit lower bounds via the Williams 2014 framework; Sipser function as explicit candidate
- **Recorded**: 2026-05-07 21:40 UTC
- **Entry ID**: `b80de3626c1d`

Statement

Encode f:{0,1}^n→{0,1} as A_f ⊆ Z_{2^n} by viewing each accepting input as its binary integer, and define the relative 3-AP density R_3(f) := |{(x,y,z)∈A_f^3 : x+z ≡ 2y (mod 2^n)}| · 2^n / max(|A_f|^3,1) (so a uniformly random subset has R_3 = 1+o(1)). Conjecture: every $ACC^0[6]$ circuit of depth d ≤ 4 and size s ≤ n^{log n} computing f satisfies R_3(f) ≥ 1 − (s/2^{n/(8d)}), so for n

≥ 12 every such circuit forces $R_3(f) \geq 0.8$; while the depth-3 Sipser function S_n (alternating AND-OR-AND with fan-in ⌈n^{1/3}⌉) satisfies $R_3(S\_n) \leq 0.6$ for all $n \geq 12$, certifying S_n $\notin ACC^0[6]$. Any single $ACC^0[6]$ sample with $R_3 < 0.8$ or any Sipser realization with $R_3 > 0.6$ falsifies the conjecture.

Rationale

R_3 measures sensitivity to additive structure under the canonical binary encoding, a coordinate the Razborov–Smolensky low-degree-polynomial machinery does not directly govern, so it offers a fresh handle inside the Williams algorithm-vs-lower-bound program. Crucially, 'low R_3' is a SPARSE property (random functions have $R_3 \approx 1$), so it is non-natural in the Razborov–Rudich sense, escaping the natural-proofs barrier listed safe by Williams while remaining constructive in time O(2^{2n}). The constructive map (truth table → integer support → 3-AP count) is fully explicit and avoids relativization/algebrization since it inspects the entire truth table rather than using oracle/polynomial-extension queries.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [FOCS.2019.00080] Why are Proof Complexity Lower Bounds Hard? - [s2:0f5716700eab6bfbf79f8f31d90a5bc579064] Computing and combinatorics : 13th Annual International Conference, COCOON 2007 Banff, Canada, July 16-19, 2007 : procee

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6cbf74e0.py", line 117, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6cbf74e0.py", line 93, in run_trial
    truth_table = evaluate_circuit(circuit, n)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6cbf74e0.py", line 63, in evaluate_circ
    output = or_gate(output, inputs[0](input_bits[inputs[1]]))
                                       ~~~~~~~~~~^^^^^^^^^^^
TypeError: list indices must be integers or slices, not function
```

Judge reasoning

The test crashed with a TypeError in evaluate_circuit (list indexed by a function), so no R_3 data was produced for either $ACC^0[6]$ samples or Sipser instances. The pre-registered support condition cannot be evaluated. | next: Fix the circuit evaluation bug (correct the gate-input tuple unpacking so the integer index, not the gate function, is used to index input_bits) and rerun the 30-seed sweep at n∈{12,14}.

Secant Variety Dimension Lower-Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry of Secant Varieties × Randomized Communication Complexity of Disjointness
- **Recorded**: 2026-05-07 21:41 UTC
- **Entry ID**: `578abb939eb0`

Statement

For a $|X|\times|Y|$ matrix M defining a partial function f, let dim(sec(M)) denote the dimension of the secant variety of the Veronese reembedding of M's row space. Then $\dim(\sec(M)) \geq \Omega(\sqrt{|X|\cdot|Y|})$ for all M corresponding to DISJ_n, with equality achieved by the standard DISJ_n matrix.

Rationale

Secant variety dimension captures the minimal number of rank-1 tensors needed to approximate M, reflecting communication complexity via the tensor rank-approximation duality. The Veronese reembedding amplifies geometric structure, making the secant dimension a proxy for the communication matrix's intrinsic complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb92e77e.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb92e77e.py", line 81, in run_trial
    dim_sec = secant_variety_dimension(M)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb92e77e.py", line 70, in secant_variet
    approx = rank_approximation(flat, rank)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb92e77e.py", line 35, in rank_approxim
    n = len(tensor[0])
        ^^^^^^^^^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

Test crashed with TypeError due to invalid len() call on integer, preventing evaluation of conjecture | next: Fix the rank_approximation function's tensor handling to resolve the TypeError and rerun tests

---

Secant Variety Dimension Lower-Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry of Secant Varieties × Randomized Communication Complexity of Disjointness
- **Recorded**: 2026-05-07 21:46 UTC
- **Entry ID**: `7181420017d3`

Statement

For a $|X| \times |Y|$ communication matrix M of the DISJOINTNESS function, the secant variety dimension of M's rank-2 tensor decomposition satisfies $\dim(\sigma_2(M)) \geq c \cdot \min(|X|,|Y|)$ for some universal constant $c>0$. This implies randomized communication complexity $R(M) \geq \Omega(\min(|X|,|Y|))$

Rationale

Secant variety dimensions capture the minimal number of rank-1 tensors needed to approximate a matrix, which directly relates to communication complexity via the tensor rank barrier. The DISJOINTNESS matrix's structure forces a linear lower bound on this dimension, revealing geometric constraints on information-theoretic protocols.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc52ee64.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc52ee64.py", line 73, in run_trial
    rank = tensor_rank(M)
           ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc52ee64.py", line 44, in tensor_rank
    B[i][j] = M[2*i+1][2*j+1]
              ~~~~~~~~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing data, preventing evaluation of conjecture support/falsification | next: Fix tensor_rank implementation to handle matrix dimensions properly

---

Lyndon Factor Width Lower-Bounds Tree-Like Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Lie algebra combinatorics (Chen–Fox–Lyndon factorization, Witt's necklace formula) × Tree-like Resolution / sub-Frege refutation length (gateway to EF lower bounds)
- **Recorded**: 2026-05-07 22:08 UTC
- **Entry ID**: `d04fdfb9056c`

Statement

Fix an unsatisfiable CNF φ on n variables and a literal ordering x_1 < ¬x_1 < x_2 < ¬x_2 < … < x_n < ¬x_n; encode each clause C as the indicator vector b(C) ∈ {0,1}^{2n}, and let L(C) be the number of factors in the unique Chen–Fox–Lyndon factorization of b(C) under this ordering. Define the proof's Lyndon-width Λ(π) = max_{C ∈ π} L(C) over all clauses (input plus derived) appearing in a tree-like Resolution refutation π of φ. We conjecture that for every minimum-length tree-like Resolution refutation π* of φ, Λ(π*) ≥ ⌈log₂(w(φ)+1)⌉, where w(φ) is the resolution width of*

π; equivalently, any refutation with all clauses having ≤ k Lyndon factors must have width < 2^k − 1, and a single derived clause with width ≥ 2^k − 1 yet ≤ k Lyndon factors falsifies the conjecture.

Rationale

Free-Lie/Lyndon decomposition is essentially absent from proof complexity, yet the Chen–Fox–Lyndon factorization is canonical and linear-time computable, providing a 'hidden non-commutative spectrum' of each derived clause. Since the size–width relation (Ben-Sasson–Wigderson) tightly couples tree-like Resolution length to width, any super-logarithmic forced growth in Lyndon-factor count would translate into super-polynomial length lower bounds, opening a fresh route to sub-Frege (and ultimately EF) lower bounds via the structure of cyclic rotations rather than via expansion or pseudo-randomness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.63s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dd85476.py", line 119, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dd85476.py", line 72, in run_trial
    for family_func, n_values, density in families:
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 3, got 2)
```

Judge reasoning

The test crashed with a ValueError during family unpacking (expected 3 values, got 2) before any refutations were generated, so no data exists to evaluate the pre-registered support or falsification conditions. | next: Fix the families tuple structure to include the density field (or unpack only 2 values) and re-run the 30×3×5 trial battery to obtain Λ vs $\lceil \log_2(w+1) \rceil$ measurements.

Polymatroid Rank Gap in Monotone DNF Depth for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone Circuit Depth for k-CLIQUE
- **Recorded**: 2026-05-07 22:09 UTC
- **Entry ID**: `289b9bdc2257`

Statement

For any monotone DNF formula computing the k-CLIQUE function on n-vertex graphs, the polymatroid rank function ρ satisfies ρ ≥ n^{1/2} · k^{1/4}, while for general monotone DNF formulas with size poly(n), ρ ≤ log n + 2 log k.

Rationale

Polymatroid rank captures submodular constraints on clause interactions. By linking it to circuit depth, we expose structural bottlenecks in representing clique-like dependencies, which are inherently non-submodular. The gap between k-CLIQUE and general DNF suggests inherent complexity in clique detection.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71a29cd3.py", line 179, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71a29cd3.py", line 104, in run_trial
    rank_k_clique = rank(A)
                    ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71a29cd3.py", line 57, in rank
    return len(gaussian_elimination(A_copy, [0]*n))
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71a29cd3.py", line 46, in gaussian_elim
    augmented[i][j] /= pivot
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. No counterexample found, and support fraction cannot be evaluated. | next: Debug gaussian_elimination to avoid division by zero and re-run experiments with validated inputs

---

Finite Plane Line Count Bounds MCSP Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Geometry × Minimum Circuit Size (MCSP)
- **Recorded**: 2026-05-07 22:33 UTC
- **Entry ID**: `f2f7b7a70710`

Statement

For a 3-SAT instance constructed from a finite projective plane of order q, the minimal circuit size required to solve MCSP is $\Theta(q^2)$. Specifically, if the instance encodes the incidence relations of lines and points in PG(2,q), then $\text{MCSP}(\Phi) \geq q^2/2$ and $\text{MCSP}(\Phi) \leq 2q^2$ for all $q \geq 2$.

Rationale

Finite geometries impose combinatorial regularity on solution spaces, which may correlate with the inherent complexity of describing these structures via circuits. The projective plane's symmetry and incidence constraints could create unavoidable computational bottlenecks in MCSP.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c76d345b.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c76d345b.py", line 70, in run_trial
    mcsp_value = branch_and_bound(cnf=clauses)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c76d345b.py", line 61, in branch_and_bo
    return backtrack({})
           ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c76d345b.py", line 57, in backtrack
    if backtrack(assignment):
       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c76d345b.py", line 57, in backtrack
    if backtrack(assignment):
       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c76d345b.py", line 57, in backtrack
    if backtrack(assignment):
       ^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 994 more times]
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

Test crashed with recursion depth error, preventing evaluation of MCSP bounds. | next: Optimize branch-and-bound algorithm with memoization or increase recursion depth limits

---

Leinster Magnitude of Hamming-Embedded Communication Matrices Lower-Bounds R(DISJ)

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Leinster metric magnitude (Euler-characteristic-style invariant of finite metric spaces via inverting the similarity matrix Z(t)=exp(-t·D)) × Randomized communication complexity of partial Boolean matrices, with DISJOINTNESS as the canonical Ω(n) target
- **Recorded**: 2026-05-07 22:35 UTC
- **Entry ID**: `c1a1a8806697`

Statement

For a 2^n × 2^n {0,1}-matrix M, embed its rows as points of the Hamming metric space H(M)=({rows of M}, d_Ham) and define the magnitude τ_t(M) := 1^T Z(t)^{-1} 1 where Z(t)[i,j] = exp(-t·d_Ham(row_i, row_j)). At the canonical scale t* = ln 2 we conjecture that (a) $\log_2$ τ_{t*(DISJ_n) = (1 ± o(1))·n as n → ∞, and (b) every 2^n × 2^n {0,1}-matrix M satisfies $\log_2$ τ_{t*}(M) ≤ C·R_{1/3}(M) for an absolute constant C ≤ 8; consequently τ_{t* is a Lean-formalizable tensor invariant providing the Ω(n) lower bound on R(DISJ_n). The conjecture is falsified by exhibiting a 2^n × 2^n matrix where $\log_2$ τ_{t*}/R_{1/3} > 8 or where DISJ_n's $\log_2$ τ_{t*} drops below 0.5n or exceeds 1.5n for any tested n.

Rationale

Magnitude captures the 'effective number of distinct points' of a metric space and is monotone under inclusions of metric subspaces, so it should detect the row-diversity that any low-communication protocol must collapse via rectangular partitions. Because rectangles in low-communication protocols force rows in the same leaf to be Hamming-close, a small protocol forces magnitude to collapse, while DISJ's rows are pairwise far apart and should sustain magnitude exponential in n. To our knowledge magnitude has never been applied as a communication-matrix invariant, despite being a single matrix-inversion away from concrete numbers.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Free Cumulant Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-07 22:46 UTC
- **Entry ID**: `1d4fc228f6cf`

Statement

For any read-twice branching program P over n variables, the maximum free cumulant of its path distribution satisfies ρ(P) = O(log n). However, for the inner product mod 2 function IP_2, ρ(IP_2) = Ω(n).

Rationale

Free cumulants capture non-commutative independence structures, which may be inherently constrained in read-twice branching programs due to their limited memory. IP_2's high complexity could manifest as unbounded free cumulants, reflecting its resistance to such structured computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_164f1942.py", line 151, in <module>
    result = run_trial(seed)
```

```
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_164f1942.py", line 119, in run_trial
    IP_2_dp = IP_2_path_distribution(n)
              ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_164f1942.py", line 90, in IP_2_path_dis
    dp = [0] * (1 << n)
         ~~~~^~~~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError before producing results, preventing validation of conjecture | next: Optimize memory usage for IP_2_path_distribution computation

---

Kronecker Coefficient Asymmetry in Symmetric Tensor Decompositions for Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Boolean Circuit Complexity of Permanent
- **Recorded**: 2026-05-07 22:58 UTC
- **Entry ID**: `c2da2c902c14`

Statement

For all n ≥ 2, the Kronecker coefficient g(λ, μ, ν) for the decomposition of Sym^n(V) ⊗ Sym^n(V) into irreducible representations is strictly greater for the permanent polynomial than for the determinant polynomial when λ, μ, ν are Young diagrams with at most n rows and columns.

Rationale

Kronecker coefficients encode the multiplicity of irreducible representations in symmetric tensor products, which are algebraic analogs of matrix multiplication. The permanent's higher symmetry compared to the determinant may manifest as asymmetric Kronecker coefficients, potentially revealing structural differences that could imply circuit complexity gaps.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_48911ad2.py", line 125, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_48911ad2.py", line 96, in run_trial
    permanent_coeff = kronecker_coefficient(lambda_, mu, nu, n)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_48911ad2.py", line 36, in kronecker_coe
    raise ValueError("Dominant weight triples must have exactly n rows/columns")
ValueError: Dominant weight triples must have exactly n rows/columns
```

Judge reasoning

Test crashed due to invalid Young diagram dimensions; cannot evaluate conjecture's validity | next: Fix Young diagram row/column constraints in kronecker_coefficient implementation

---

Cubical Euler Characteristic Bounds DNF-Min for Symmetric Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Cubical algebraic topology (Euler characteristic of subcomplexes of the n-cube) × DNF-min meta-complexity (gap-DNF-MCSP) for symmetric Boolean functions
- **Recorded**: 2026-05-07 23:11 UTC
- **Entry ID**: `ed9920e4ad50`

Statement

For every symmetric Boolean function f:{0,1}^n→{0,1} (n≥2), determined by its on-weight set R_f:={k∈{0,…,n}: f(any weight-k input)=1}, define the cubical complex K_f as the union of all faces C of the n-cube on which f≡1, and let χ(K_f) be its Euler characteristic, computed by the closed form χ(K_f)=Σ_{d=0}$^{\{n\}(-1)}$d·C(n,d)·Σ_{w_0:[w_0,w_0+d]⊆R_f} C(n-d,w_0). Then DNF-min(f) ≥ |χ(K_f)|, with equality whenever R_f consists only of isolated weights (so the bound is saturated by parity, where both equal 2^{n-1}). One symmetric f with DNF-min(f) < |χ(K_f)| falsifies the conjecture.

Rationale

Each minterm of a DNF for f is a face of K_f, so χ(K_f) measures the inclusion–exclusion residue across all such faces; a small DNF has few subcubes available to cancel into χ, forcing |χ| to be small. Restricting to symmetric f (a 2^{n+1}-size family) sidesteps the natural-proofs barrier — the discriminating property is not 'large' on the full 2$^{\{2}$n} function space, evading Razborov–Rudich pseudo-random-function obstructions while still using a constructive, poly(2^n)-computable invariant. The bridge between cubical Euler characteristic and exact DNF-minimization for the symmetric class appears nowhere in the standard literature on MCSP, set-cover, or AC^0 lower bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2306.14401v1] On the distribution of sensitivities of symmetric Boolean functions - [1805.08177v4] A Polynomial Time Delta-Decomposition Algorithm for Positive DNFs - [0808.0684v1] 9-variable Boolean Functions with Nonlinearity 242 in the Generalized Rotation Class - [2605.00705v1] Cohomological properties of the Vietoris–Rips Complex of a Hypercube Graph - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e3099eab.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e3099eab.py", line 116, in run_trial
    if all(f[binary_rep[:j] + '1'] for j in range(len(binary_rep))):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e3099eab.py", line 116, in <genexpr>
    if all(f[binary_rep[:j] + '1'] for j in range(len(binary_rep))):
```

```
          ~^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'function' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError ('function' object is not subscriptable) before producing any data, so neither support nor falsification conditions were evaluated. | next: Fix the test harness bug (treat f as a callable, not a dict — replace f[binary_rep[:j]+'1'] with f(binary_rep[:j]+'1') or precompute f as a dict) and rerun the pre-registered protocol.

Random Matrix Spectral Norm and SOS Refutation Degree for 3-CNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Refutation Degree for 3-CNF
- **Recorded**: 2026-05-07 23:12 UTC
- **Entry ID**: `898d413a1fde`

Statement

For a random 3-CNF formula with n variables and m clauses, the spectral norm of its Gram matrix G (defined by G_{i,j} = ⟨var_i, var_j⟩ where var_i is the vector of clause participation for variable i) satisfies ||G|| = Θ(√(m/n)) and the SOS refutation degree required to prove unsatisfiability is Θ(log n / log ||G||).

Rationale

Random matrix theory captures the statistical structure of clause-variable interactions in CSPs, while SOS refutation degree measures algebraic complexity. The spectral norm governs the 'effective dimension' of the problem, suggesting a trade-off between problem structure and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e76d1b16.py", line 84, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e76d1b16.py", line 63, in run_trial
    norm = spectral_norm(matrix)
           ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e76d1b16.py", line 45, in spectral_norm
    v = [Fraction(1, math.sqrt(n))] * n
         ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

Judge reasoning

Test crashed due to TypeError in Fraction initialization, preventing data collection | next: Fix Fraction argument types in spectral_norm function and re-run tests

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Real Radical Rank and SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree of Max-CUT
- **Recorded**: 2026-05-07 23:34 UTC
- **Entry ID**: `937359348344`

Statement

For random Max-CUT instances on n variables, the real radical rank of the moment matrix M_d is $\Theta(\sqrt{n})$ when $d \geq \Omega(\sqrt{n})$. If any instance has real radical rank $o(\sqrt{n})$ for $d = \Theta(\sqrt{n})$, then the SOS degree required to achieve 0.878-approximation is $\omega(\sqrt{n})$.

Rationale

Real radical rank captures the minimal number of constraints needed to certify non-negativity, which directly relates to SOS degree. Lower rank implies fewer constraints, requiring higher degree to maintain approximation guarantees. This bridges algebraic geometry's structure with SOS hierarchy complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90a52e41.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90a52e41.py", line 62, in run_trial
    rank = real_radical_rank(M_d)
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90a52e41.py", line 44, in real_radical_
    _, R = qr_decomposition(A)
           ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90a52e41.py", line 34, in qr_decomposit
    Q[k][k] = R[k][k] / norm
              ~~~~~~~~^~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero error, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix numerical instability in QR decomposition or implement alternative rank computation for singular matrices

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Lorentzian Defect of Tseitin Moment Polynomial Lower-Bounds SOS Degree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Lorentzian polynomials and Hodge-theoretic combinatorics (Brändén–Huh 2020 signature theory) × Sum-of-Squares (Lasserre) refutation degree of Tseitin Max-CUT on expanders
- **Recorded**: 2026-05-07 23:41 UTC
- **Entry ID**: `2714b4a401ee`

Statement

For a d-regular graph $G=(V,E)$ with odd parity $\sigma:V\to\{0,1\}$ (so the Tseitin formula $T(G,\sigma)$ is unsatisfiable), define the degree-2 Tseitin moment polynomial $P_{G,\sigma}(x)=\Sigma_{e,f\in E}\, c_{e,f}(\sigma)\cdot x_e x_f$ with $c_{e,f}(\sigma)=|\{v\in e\cap f:\ \sigma(v)=1\}|$ if e,f share a vertex and 0 otherwise. Let $\delta(G,\sigma)$:=(number of strictly positive eigenvalues of the Hessian $H_{P_{G,\sigma}}(1)$)−1 be the Lorentzian defect. Conjecture: for every d-regular Ramanujan expander G ($\lambda_2(G)\leq 2\sqrt{(d-1)}$) and every odd σ, $\delta(G,\sigma) \geq \lceil\alpha\cdot|E|/d^2\rceil$ for an absolute constant $\alpha\geq 1/8$, and the resolution width $w(T(G,\sigma))$ (a known SOS-degree lower bound) satisfies $w(T(G,\sigma)) \geq 2+2\delta(G,\sigma)$; a single (G,σ) violating either inequality refutes the conjecture.

Rationale

Lorentzian polynomials are precisely those whose Hessian has signature (1,m−1) at all positive points, a rigid algebraic property tied to Hodge-Riemann relations on matroids. Expansion forces the Tseitin parity obstruction to scatter across many edge-pairs, so the contribution matrix should have many positive eigendirections—a defect that should reappear as algebraic hardness in SOS, since SOS reductions preserve the moment-matrix signature up to bounded distortion. This connects Brändén–Huh signature combinatorics, which (to our knowledge) has not been used in proof complexity, to the open Tseitin/SOS lower-bound program.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [2304.08399v3] Dually Lorentzian Polynomials - [2206.02759v4] Polynomials with Lorentzian Signature, and Computing Permanents via Hyperbolic Programming - [1209.2976v3] Sums of squares of polynomials with rational coefficients

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_543310d6.py", line 181, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_543310d6.py", line 112, in run_trial
   ramanujanish_graphs = [G for G in ramanujanish_graphs if is_ramanujanish(G, d, lambda_2_threshold)
                                                            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_543310d6.py", line 24, in is_ramanujani
    degree_sum = sum(degree for _, degree in G.items())
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

TypeError: unsupported operand type(s) for +: 'int' and 'set'
```

Judge reasoning

The test crashed with a TypeError in is_ramanujanish before producing any data, so no trials were evaluated against the pre-registered criterion. Per the rules, a crash yields INCONCLUSIVE. | next:

Fix the graph representation bug in is_ramanujanish (sum degrees from an adjacency-set dict via len(neighbors), not by adding sets) and rerun the pre-registered grid.

Persistent Homology Barcodes and Resolution Proof Size for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent Homology × Resolution Proof Size
- **Recorded**: 2026-05-07 23:53 UTC
- **Entry ID**: `a97487bf0a56`

Statement

For a Tseitin formula derived from a graph G, the maximum persistence in the 1-dimensional persistent homology of its resolution graph is inversely proportional to the size of the minimal resolution proof. Specifically, max_persistence(G) ≥ $\log_2$(size_of_proof) / n.

Rationale

Persistent homology captures the topological structure of the resolution graph's connectivity, where long-lived homology features (high persistence) may correspond to redundant proof steps. This bridge could reveal geometric constraints on proof complexity, leveraging the interplay between algebraic topology and combinatorial proof systems.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_472f26be.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_472f26be.py", line 79, in run_trial
    chain_complex[edge[1]][edge[0]] = Fraction(1)
    ~~~~~~~~~~~~~~^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing data, preventing evaluation of the conjecture. | next: Debug the IndexError in the test code, particularly the chain_complex[edge[1]][edge[0]] assignment

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Free Cumulant Rank Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-07 23:58 UTC

- **Entry ID**: `076ad72eacfc`

Statement

For any read-twice branching program P with size S, the rank of its free cumulant matrix κ(P) satisfies rank(κ(P)) = O(log S). For the IP_2 trivial BP (constant-depth, size 2^n), rank(κ(P)) = Ω(n).

Rationale

Free cumulants capture non-commutative independence structures that may distinguish read-twice BPs (with limited state reuse) from IP_2's inherently high-dimensional correlations. The rank gap could expose algebraic constraints on BP expressiveness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.09s`


```
'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': False, 'counterexamp
TRIAL: {'metric_name': 'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds
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TRIAL: {'metric_name': 'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'cumulant_rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds
RESULT: FALSIFIED counterexample="rank exceeds expected" first_failing_seed=11
```


Judge reasoning

Test failed to scale with n=1, violating IP_2's required n=Ω(log S). Metric_value=40 exceeds impossible IP_2_rank=36 for n=1. | next: Test with n=5 and S=2^5=32 to validate IP_2_rank=Ω(n) scaling